

DANIEL BENJAMIN

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Hardware Development Engineer Intern/Co-op | Available May 2027 for 4 or 8 months

EDUCATION

University of British Columbia

Expected Graduation: May 2028

Bachelor of Applied Science – Mechanical Engineering

ROBOTICS & HARDWARE DEVELOPMENT HIGHLIGHTS

- Designed, prototyped, and tested a [soft-pneumatic robotic hand](#), improving grasping performance by 25% over prior iterations
- Designed and manufactured an [FSAE chassis](#) in SOLIDWORKS; wrote Python load-case tools for ANSYS torsional-rigidity simulations
- Designed, 3D printed, and validated multiple [electronics enclosures](#) against IPX5 water-ingress requirements
- Automated mechanical CAD workflows with AutoLISP, reducing drafting time by over 30% and saving 7+ hours per week

SELECTED EXPERIENCE

Chassis Team Lead (prev. Team Member), UBC Formula Electric

Sep 2023 – Present

- Lead a 7-person subteam through design, fabrication, and integration milestones for the 2026 competition vehicle
- Plan and execute carbon- and glass-fiber composite layups for vehicle body panels and aerodynamic components
- Deployed an SVN server and [custom SOLIDWORKS integration](#) to control design revisions across the team
- Built a [3D-printer operations dashboard and Slack bot](#) for job submission, live status, and reservations
- Write design documentation and manufacturing SOPs to standardize builds and transfer technical knowledge

Mechanical Co-op, Avalon Mechanical

May 2025 – Aug 2025

- Developed AutoLISP automation scripts that reduced CAD drafting time by over 30%, saving 7+ hours per week
- Produced AutoCAD drawing revisions, redlines, and technical documentation for tender and construction packages
- Performed pressurization analysis and fan sizing in Excel and TRACE to meet NFPA and provincial codes
- Validated fire sprinkler hydraulic systems using Elite Software FIRE for code-compliant designs

Undergraduate Research Assistant, UBC MEMS Lab

May 2024 – Feb 2025

- Analyzed 15+ tendon-driven prosthetic hands to translate successful mechanisms into design requirements
- Rapidly prototyped and tested iterations on a custom multi-extrusion 3D printer running RepRapFirmware
- Designed a silicone return mechanism around actuator force limits and integrated it into the robotic hand
- Developed a MATLAB- and Arduino-based demonstration platform that reproduced human hand gestures

Volunteer, Victoria Hand Project

July 2022 – Present

- Assemble, troubleshoot, and repair low-cost 3D-printed prosthetic arms with voluntary open/close functionality
- Improved assembly workflows and documented the revised process for future volunteers

PROJECTS

AI-Driven Predictive Maintenance System

github.com/danielrbenjamin/manu465-project

- Built a [TensorFlow model in Python](#) to predict maintenance needs from smart-manufacturing equipment data

SKILLS

Hardware: SOLIDWORKS, AutoCAD, ANSYS Mechanical, FEA, rapid prototyping, hardware testing, composites, machining, 3D printing, FMEA

Software: Python, MATLAB, C, Arduino, AutoLISP, JavaScript, Git, SVN, Docker, AWS, Jira